TEXORA · SESSIONAL NOTES · STUDY SHEET

Preparation of Production Pattern of a Basic Shirt

Drafting the production pattern of a basic shirt from a measurement chart, step by step: construction frame, back, front, sleeve, collar and cuff — then allowances, notches, grain lines and the checking that makes a draft into a production pattern.

TE-2208 · APPAREL MANUFACTURING TECHNOLOGY-1 SESSIONAL · STUDY 3 · LEVEL 2 TERM 2

1. Aim and place of the study

Aim: to prepare the complete production pattern of a basic shirt from a given measurement chart, following the drafting sequence used in the pattern room. The principles — drafting order, allowances, grain, walking the seams — are set out in TE-2207 Topic 3; this study is their bench application to one garment, in one size, ready to be examined piece by piece in Study 4.

2. Materials and the measurement chart

Materials: pattern paper, L-square and long ruler, measuring tape, French curve, pencil and eraser, notcher, awl, pattern shears, and weights to hold the paper flat.

Measurement chart (example — men's basic shirt, size M, garment values in cm): chest 107 · body length 73 · shoulder width 46 · sleeve length 61.5 · collar (neck) size 40. These are the size-M values of the specification studied in TE-2207 Topics 2–3, used unchanged so that all sheets of this course agree. From them the draft derives: half chest = 53.5 cm; quarter chest = 26.8 cm; half shoulder = 23 cm.

> **READING THE CHART**
>
> A pattern is drafted in half (or quarter) whenever the garment is symmetric: the front draft represents one half of the body width, built on the centre-front line, and the back draft likewise on the centre-back line. Lengths are drafted full. Every derived value must be written on the draft as it is used — the next student, or the examiner, reads the pattern through its numbers.

3. Step-by-step drafting

3.1 The construction frame and the back

1. Draw the vertical centre-back line and, square to it at the top, the neck/shoulder line. Mark the full body length 73 cm down the centre line to the hem line.
2. Square out the chest line at the scye (armhole) depth below the shoulder line — for this draft about 26 cm (proportional to chest; the exact depth follows the drafting system in use and is recorded on the pattern).
3. On the chest line mark the quarter chest 26.8 cm from the centre line; this sets the side-seam position of the half pattern.
4. Shape the back neckline: neck width of about one-fifth of the collar size (40 ÷ 5 = 8 cm) from the centre line, curving down about 2 cm at the centre back.
5. Draw the shoulder seam from the neck point to the shoulder point (half shoulder 23 cm from the centre line), with its normal downward slope; shape the armhole from the shoulder point with the French curve down to the side seam at the chest line.
6. Draw the yoke line across the back, roughly 9–10 cm below the shoulder line at the centre, and mark the pleat positions below it. Complete the side seam and the hem (slightly curved).

3.2 The front

1. Use the same frame dimensions so front and back share length, chest and side-seam geometry.
2. Deepen the front neckline: the front neck curve drops about 7–8 cm below the shoulder line at the centre front for the collar band to sit in.
3. Add the button placket extension — about 2 cm grown-on beyond the centre-front line — and mark the button positions down it (first button below the band, then evenly spaced).
4. Shape the front armhole slightly deeper than the back's, and mark the chest pocket position with drill holes (its top edge about 20 cm below the shoulder line, measured on the garment during the study).

3.3 The sleeve

1. The sleeve is drafted from the armhole: measure the front and back armhole curves of the draft with the tape on edge; their sum (about 50 cm for this size) is the length the sleeve cap must match, with a small designed cap ease of about 1–1.5 cm.
2. Draw the sleeve centre line to the full sleeve length 61.5 cm (this includes the cuff width, which is drawn separately and deducted when the pieces are separated).
3. Set the cap height from the armhole depth (about half the scye depth in this system), the biceps width from the chest proportion, and shape the cap curve so its length equals the armhole plus the cap ease.

4. Shape the underarm seams from biceps to cuff opening (cuff opening about 24 cm — wrist plus movement and overlap), and mark the sleeve placket position on the back underarm seam.

3.4 Collar, band and cuff

1. Measure the front and back neckline curves on the draft; their sum is the neckline length the band must follow (it will equal the collar size 40 cm when the draft is correct — walking check in §5).
2. Draft the band as a gently curved strip of that length, about 3 cm wide; draft the fall over it, about 4.5–5 cm wide at the centre back, shaped to the collar style (points at the front ends).
3. Draft the cuff as a rectangle: length = cuff opening 24 cm plus button/buttonhole overlap about 2.5 cm; width about 6–7 cm per style. Mark button and buttonhole positions.

4. From draft to production pattern

The drafted lines are the **net (sewing) lines**. Around each piece add the allowances of TE-2207 Topic 3, which this course uses unchanged:

WHERE	ALLOWANCE
Most seams (shoulder, armhole, yoke, placket edges)	1 cm
Side seams and centre back of yoke (adjustment seams)	1.5 cm
Enclosed curved seams — collar, band, cuff edges	0.6–0.75 cm
Shirt hem	2.5 cm (within the 2–2.5 cm range of the theory sheet)

- **Grain line:** an arrowed line on every piece, parallel to the centre line of the body pieces and to the sleeve centre line; collar fall on the straight grain, band likewise.
- **Notches:** single notch on front edges, double notch on back edges (the standing convention of TE-2207); notches at sleeve cap (front/back identification), at yoke joining points, and at the cuff and placket positions.
- **Drill holes:** pocket position, button positions on the placket, dart/pleat ends.
- **Information block on every piece:** style (basic shirt), piece name, size M, cut quantity (front ×2, back ×1, sleeve ×2, cuff ×2, band ×2 layers with interlining, fall ×2 with interlining, yoke ×1 doubled, pocket ×1).

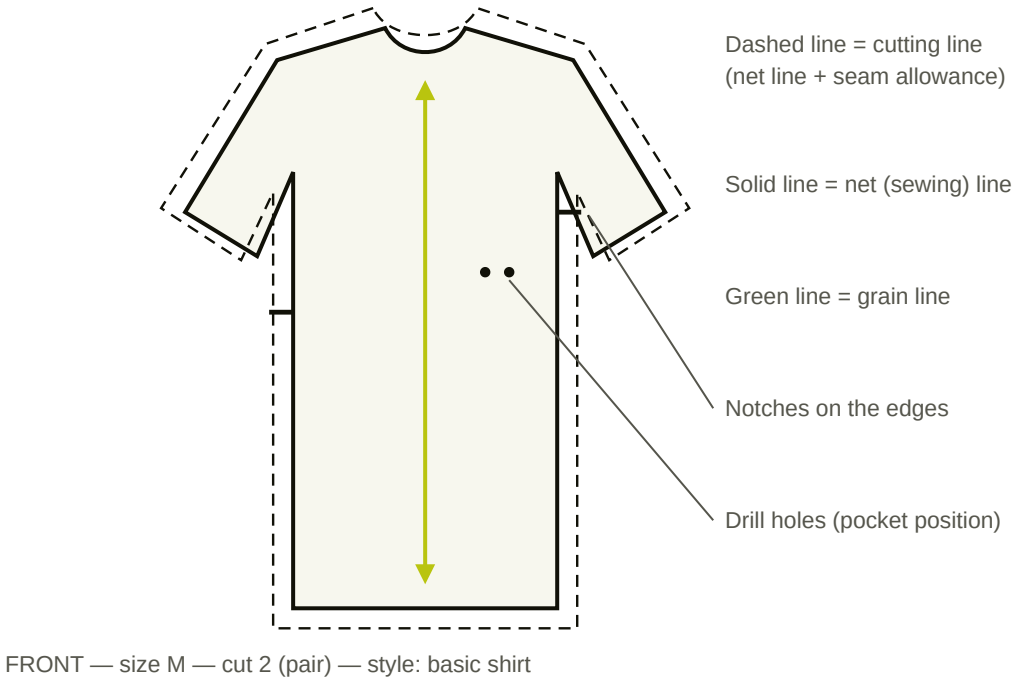


Figure 1 — The front piece as a finished production pattern: net line, cutting line outside it, grain line, notches and drill marks, and its information written on the piece.

5. Checking before the pattern is released

- **Walk the seams:** front and back side seams measured against each other; sleeve cap walked around the armhole (cap = armhole + about 1–1.5 cm ease); band walked along the neckline (must equal 40 cm).
- **Check the set:** every piece of §2's component list present, pairs truly mirrored, interlining pieces included for collar, band and cuff.
- **Check the measures:** finished (net) dimensions re-measured on the pattern and compared with the chart: half chest ×4 = 107, length 73, sleeve 61.5 less cuff width plus cuff = 61.5.

6. Precautions

- Keep the L-square honest: one un-squared frame line tilts every line built on it.
- Add allowances only after the net shape is checked — allowances faithfully copy errors as well as shapes.
- Cut the paper pattern on the cutting line with long, clean strokes; a nicked paper edge becomes a nicked fabric edge in every lay.
- Write the information block before separating the pieces; unlabelled pieces separated from the set cannot be identified with certainty later.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. List the drafting order for the shirt pattern set, and state why the front is drafted on the same frame as the back.
2. From the chart, compute the quarter chest and the neck width used for the back neckline, showing the divisions used.
3. State the seam allowances used on (a) the side seam, (b) the collar edge, (c) the hem, and the reason the collar edge takes less than the side seam.
4. The band is walked along the neckline and measures 41.6 cm against a collar size of 40. What does this tell you, and what must be corrected before cutting?
5. What cap ease is allowed between sleeve cap and armhole, and what happens in sewing if the cap is exactly equal to the armhole or far larger?
6. Name the markings, other than the outline, that a production pattern piece must carry.
7. Why is the pattern drafted in half for the body pieces but the sleeve length drafted full?

Answer key

1. Frame and back → front on the shared frame → sleeve from the armhole → collar/band from the neckline → cuff and pocket. Front and back share the frame so that chest, length and side-seam geometry agree by construction.
2. Quarter chest = 107 ÷ 4 = 26.8 cm (drafted from the centre line). Back neck width = collar ÷ 5 = 40 ÷ 5 = 8 cm.
3. (a) 1.5 cm (adjustment seam). (b) 0.6–0.75 cm. (c) 2.5 cm hem allowance. The collar edge is enclosed and curved: a wide allowance would add bulk and prevent the edge from lying smoothly when turned.
4. The neckline of the draft is 1.6 cm longer than the collar it must carry — front and back neck curves were over-shaped. The neckline (or the band draft) must be corrected and re-walked; cut as it stands, the collar would not fit the neck opening.
5. About 1–1.5 cm of ease. With zero ease the sleeve sets flat and strains; with far too much, the cap cannot be eased in without puckers and tucks at the armhole.
6. Grain line, notches (single front / double back), drill holes for internal positions, and the information block: style, piece name, size, and number to cut.
7. The body is symmetric about its centre lines, so a half draft generates the whole piece on the fold or as a pair; the sleeve length is a single one-dimensional measure along the arm, which is simply laid out in full along the sleeve centre line.